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ACADEMIC VAULT

UNIZIK FRESHER EDITION | 2026

GST 103: Philosophy and Logic - Arguments, Fallacies and Clear Thinking

Learn to tell a real argument from noise, and to build one that survives cross-examination

GST / COMPULSORY

GST & Compulsory Courses

General Studies (GST / CED)
GST Directorate | 100L

Written for students, by students.

Practical notes, worked examples, practice questions and a revision plan.

Study smarter. Ask better questions. Pass with understanding.

Independent study support | Not an official UNIZIK publication

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Why this pack exists

Philosophy and Logic is where many science and management students switch off - until they meet the objective questions. This pack explains validity, soundness, syllogisms and informal fallacies in plain language, with everyday UNIZIK examples and a fallacy-spotting drill.

This is a student-written study companion, not a reproduced lecture handout or a leaked examination paper. Use it to organise the topics you are taught, practise the kind of thinking your course rewards, and walk into class with better questions.

What you will be able to do

- Distinguish an argument from a statement, explanation or quarrel.
- Test an argument for validity and soundness without confusing the two.
- Recognise the major informal fallacies in speeches, adverts and social-media arguments.
- Translate simple arguments into standard form and evaluate categorical syllogisms.

How to use this pack

1. Skim the whole pack once in 20 minutes so you know what is inside.
2. Work through one section at a time with your lecture note and recommended text beside you.
3. Attempt the practice questions before reading the answers. Mark what you missed.
4. Use the final checklist to plan your revision week, one tick per study session.

> **Confirm your official course outline**
> Course codes, credit units and outlines can differ by department and session. Treat this pack as supplementary support and confirm the current outline with your lecturer, course rep or departmental handbook before an assessment.

What philosophy is doing

Philosophy is not a collection of opinions. It is the disciplined practice of asking what a claim means, what it assumes, and what would count as evidence for it.

The main branches you will meet are metaphysics (what exists), epistemology (what knowledge is and how we get it), ethics (how we should act), aesthetics (beauty and art) and logic (the rules of correct reasoning).

For examinations, logic carries the most marks because it is teachable and testable. The other branches matter for definitions and short-answer questions.

- Metaphysics: reality, mind, causation, freedom.
- Epistemology: belief, truth, justification, scepticism.
- Ethics: right action, virtue, consequences, duty.
- Logic: arguments, validity, proof, fallacy.

Arguments: the basic unit

An argument is a set of statements where some, the premises, are offered as support for another, the conclusion.

'I am tired, therefore I will sleep' is an argument. 'I am tired because I slept late' is an explanation. 'You are wrong because you are annoying' is an insult wearing an argument's clothes.

Premises answer 'what is the evidence?' The conclusion answers 'what follows?' Indicator words help: therefore, hence, so, thus and consequently usually introduce conclusions; because, since, given that and for usually introduce premises.

> **Validity versus soundness**
> Validity is about structure: if the premises were true, the conclusion would have to be true. Soundness is validity plus actually true premises. An argument can be valid and unsound, but never sound and invalid.

Three patterns worth memorising

Most exam syllogisms use a small number of forms. Learn the patterns and you can evaluate them quickly.

- Modus ponens: If P then Q. P. Therefore Q. Valid.
- Modus tollens: If P then Q. Not Q. Therefore not P. Valid.
- Affirming the consequent: If P then Q. Q. Therefore P. Invalid - other causes of Q may exist.
- Denying the antecedent: If P then Q. Not P. Therefore not Q. Invalid - Q may still happen for another reason.
- Disjunctive syllogism: Either P or Q. Not P. Therefore Q. Valid when the options are genuinely exclusive.

Categorical syllogisms in plain language

A categorical syllogism has two premises and a conclusion built from three terms: major, minor and middle.

Example: All students write examinations. Ada is a student. Therefore Ada writes examinations. The middle term, 'students', links the other two terms.

Watch the quantifiers: all, no, some, some not. 'Some' means at least one, not most. A conclusion cannot introduce a stronger claim than the premises support.

- All A are B; all B are C; therefore all A are C. Valid.
- No A are B; some C are A; therefore some C are not B. Valid.
- All A are B; some B are C; therefore some A are C. Invalid - the C may be among the Bs that are not A.

Informal fallacies: the ones you will actually meet

A fallacy is a pattern of reasoning that looks persuasive but fails. Learning the names is useful; learning to explain why each fails is what earns marks.

- Ad hominem: attacking the person instead of the argument.

- Appeal to authority: citing a famous person outside their expertise.
- Appeal to popularity: 'everybody does it' does not make it true or right.
- False dilemma: presenting two options when more exist.
- Slippery slope: claiming one small step must lead to an extreme outcome without evidence.
- Straw man: attacking a weaker version of an opponent's position.
- Circular reasoning: assuming the conclusion in the premises.
- Hasty generalisation: concluding from a tiny or biased sample.
- Post hoc: assuming that because B followed A, A caused B.

> **In the exam**
> When asked to identify a fallacy, name it and write one sentence explaining the reasoning error. A name without an explanation is half an answer.

PRACTICE, NOT PREDICTION

Check your understanding

These questions are written for this pack to test the ideas above. They are not past examination questions and do not predict what your lecturer will set. Answer first, then compare.

1. Is this valid or sound: 'All birds fly. Penguins are birds. Therefore penguins fly.'?

Answer: The form is valid, but the argument is unsound because the first premise is false. Soundness requires true premises as well as valid structure.

2. Identify the fallacy: 'Don't listen to his advice on saving money - he failed his first business.'

Answer: Ad hominem. The speaker attacks the person rather than addressing the financial advice.

3. What is the difference between an explanation and an argument?

Answer: An argument offers evidence for a conclusion. An explanation shows why something is the case, assuming the fact is already accepted.

4. Test the form: If it rains, the ground is wet. The ground is wet. Therefore it rained.

Answer: Invalid - affirming the consequent. The ground could be wet for another reason, such as a burst pipe.

5. Name the fallacy: 'Either you support the strike or you hate students.'

Answer: False dilemma. It presents only two options when other positions are possible.

6. Why is 'some' different from 'most' in logic?

Answer: 'Some' means at least one. 'Most' means more than half. A conclusion using 'most' cannot be inferred from premises that only claim 'some'.

Your 7-step action plan

Tick each step only when you have done it. A tick you cannot defend is worse than an empty box.

- [ ] Write definitions of validity and soundness from memory, each with one example.
- [ ] Build your own valid modus ponens and invalid affirming-the-consequent argument on a campus topic.
- [ ] Collect ten headlines or adverts and name the fallacy in at least five.
- [ ] Draw a table of the six fallacies you find hardest and write a one-line rebuttal for each.
- [ ] Take a past objective question on logic and explain why each wrong option is wrong.
- [ ] Have a friend give you a one-paragraph argument and evaluate its premises and conclusion.
- [ ] Teach the difference between validity and soundness to someone in five minutes.

Keep going

One pack will not carry a semester. Pair this with your lecture notes, tutorials, and the past questions your department actually releases. If a topic still feels unclear, take a specific question to your lecturer or course rep - "I do not understand the difference between X and Y" gets a better answer than "I do not understand this course".

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YOUR WORKSPACE

Notes, questions and things to ask

Use this page in class or during revision. Write questions you could not answer instead of copying slides.

Three ideas I can now explain without notes:

A question I still cannot answer:

For my lecturer or course rep:

Next revision action and date: